MAS508-G Control Theory
Lecture 3: State Space Design

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L3. State Space Design

a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – LQG

Textbook, “Control Systems Engineering”, Chapters 4 &12

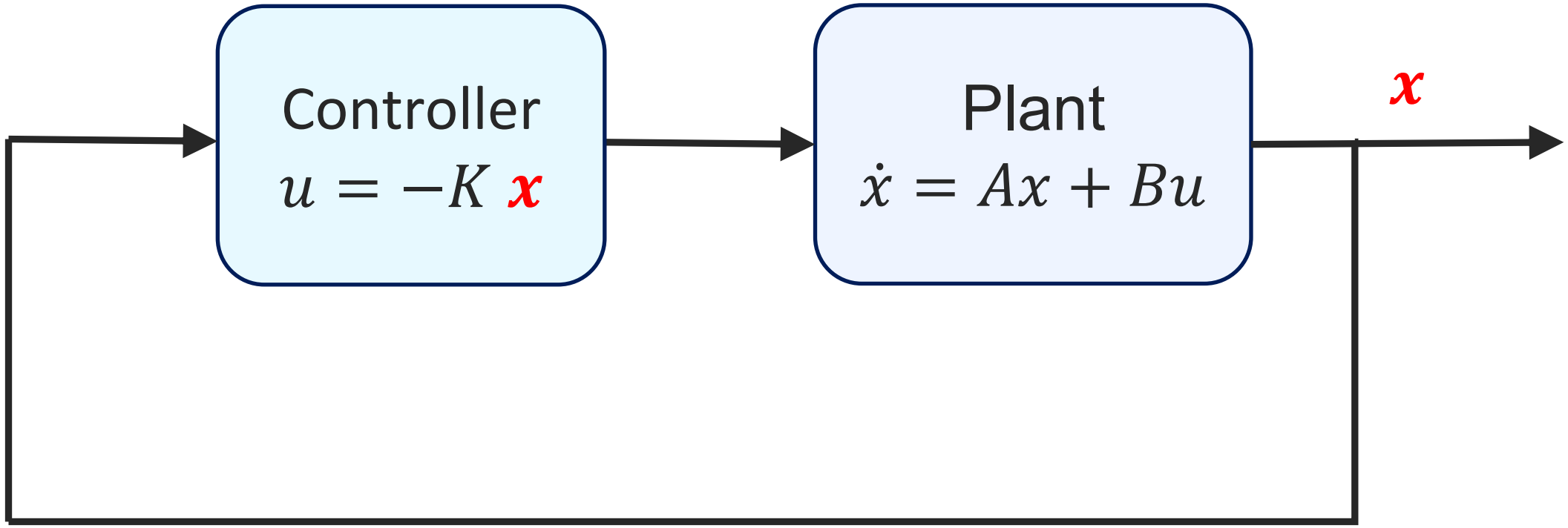
L3f. Output Feedback Control - Controller with Observer

Output Feedback

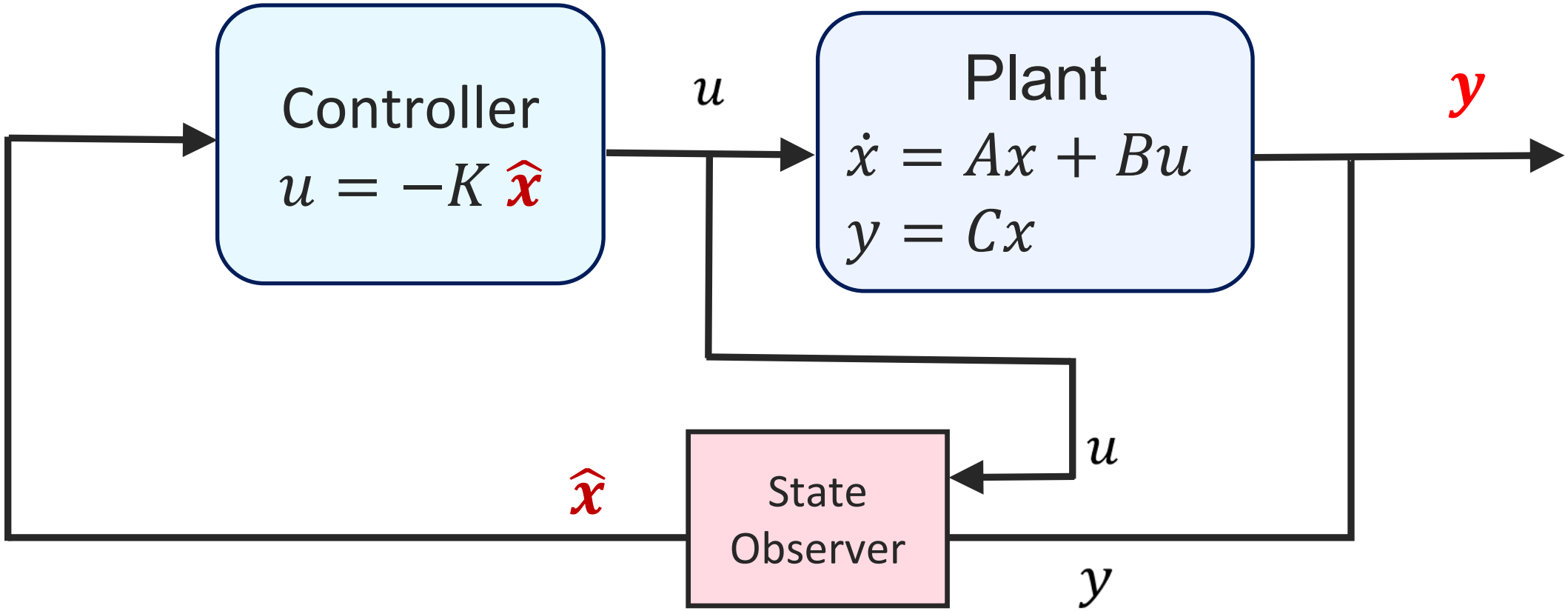
- In practice for many system, not all the states of a system are being measured

- However, if system is *observable*, then you can design a *state observer* to estimate the state. This is called *output feedback. (This is introduced in the Observer Design)*

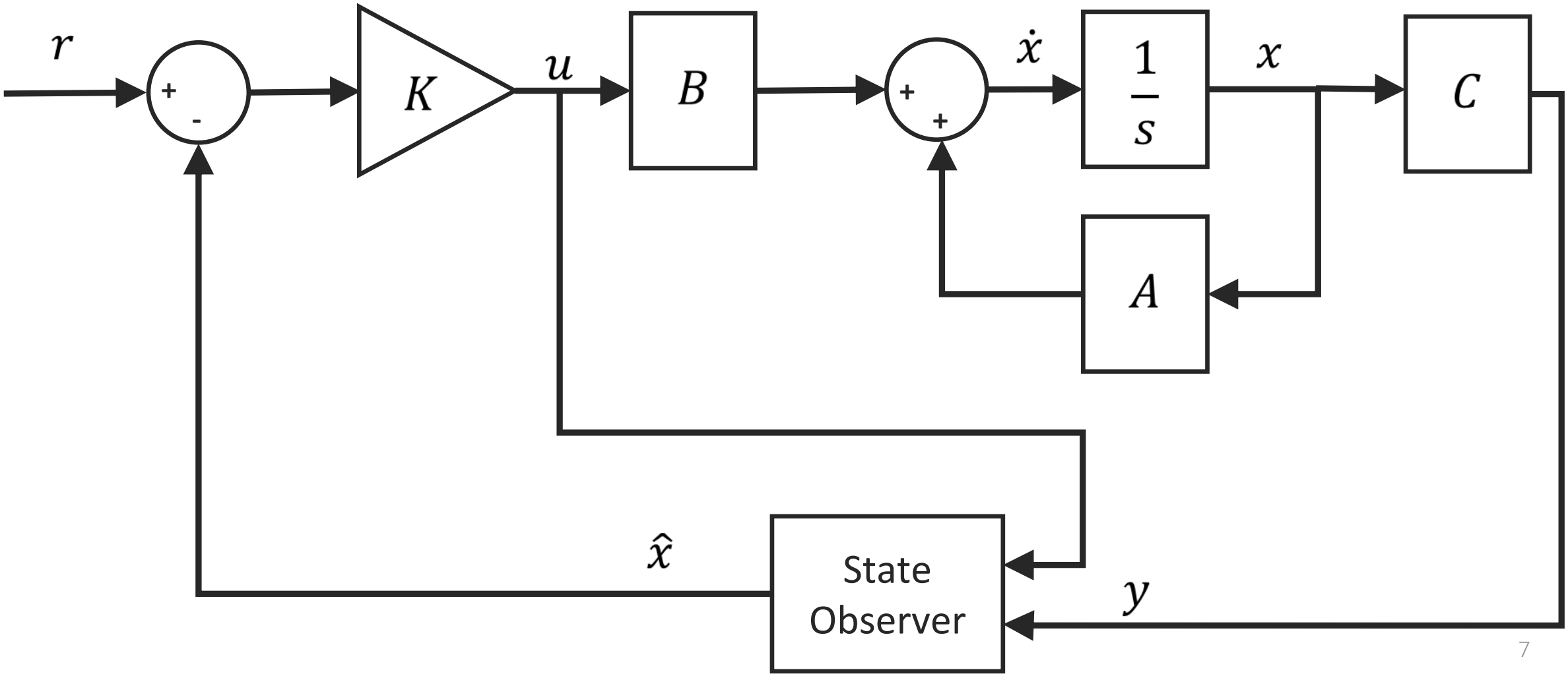
State feedback control- Full state



Output feedback control



Output Feedback with Observer



Closed-loop control problem

- We now consider the ***output feedback control*** problem for an ***observable system*** described by

$$\dot{x} = Ax + B\,u$$
$$y = Cx$$

with a ***closed-loop observer*** described by

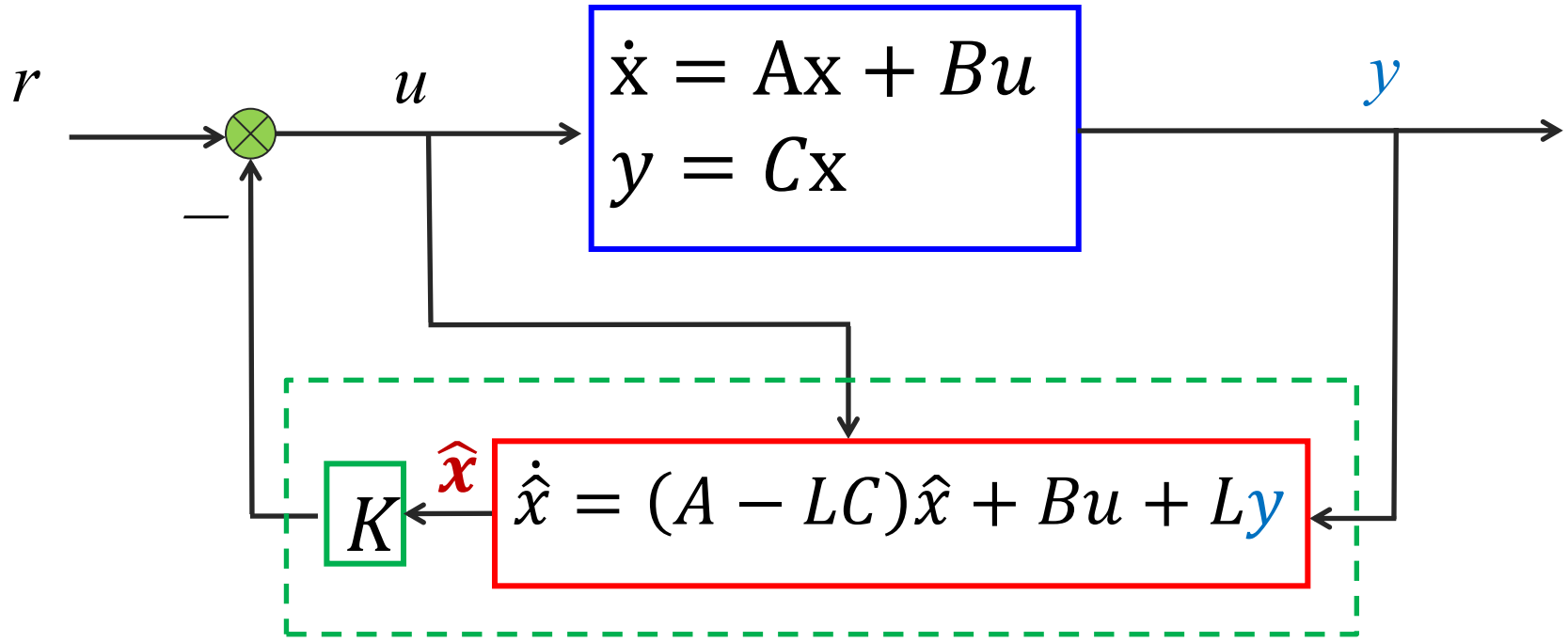
$$\dot{\hat{x}} = (\boldsymbol{A} - \boldsymbol{L}\,\boldsymbol{C})\hat{x} + \boldsymbol{B}u + \boldsymbol{L}\,y$$

and feedback control

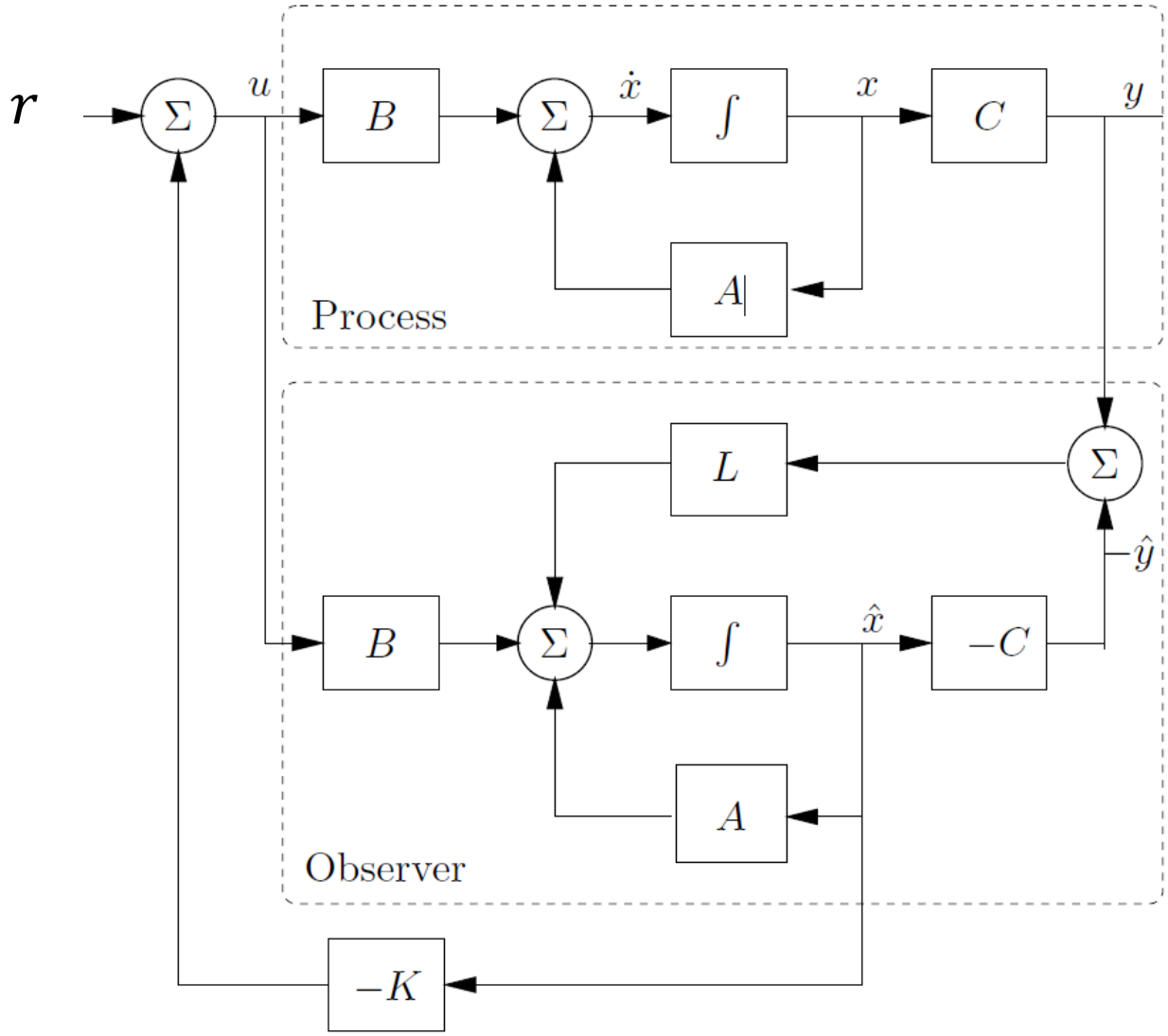
$$u = -K\hat{x} + r$$

- Note that the $\hat{x}$ is used in feedback control rather than $\boldsymbol{x}$ ***which is (assumed) unavailable for feedback***.

- **Block diagram of output feedback control system**



Output Feedback Control



Feedback control

$$u = -K\hat{x} + r$$

State observer

$$\dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly$$

- The observer and the control can be combined to yield

$$\dot{x} = Ax + Bu = Ax + B(-K\hat{x} + r) = Ax - BK\hat{x} + Br$$
$$y = C\,x$$
$$\dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly$$
$$= (A - LC)\hat{x} + B(-K\hat{x} + r) + L\,Cx$$
$$\dot{\hat{x}} = LC\,x + (A - LC - BK)\hat{x} + Br$$

which can be written in partitioned form as

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & -BK \\ LC & A - LC - BK \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} r$$
$$y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}$$

Output feedback loop control problem

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & -BK \\ LC & A - LC - BK \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} r$$

$$y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}$$

- There are now $2n$ state variables in the feedback system, n due to the system and n due to the observe

- Apply a transformation of the variables $\begin{bmatrix} x \\ \tilde{x} \end{bmatrix} = \begin{bmatrix} x \\ x - \hat{x} \end{bmatrix}$
- Compute the derivative $[\boldsymbol{x}, \tilde{\boldsymbol{x}}]^T$

$$\begin{bmatrix} \dot{x} \\ \dot{\tilde{x}} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ \tilde{x} \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} r$$

$$y = [C \quad 0] \begin{bmatrix} x \\ \tilde{x} \end{bmatrix}$$

- Compute the transfer function of the closed-loop system

$$\frac{Y(s)}{R(s)} = [C \quad 0] \left(sI_{2n} - \begin{bmatrix} A - BK & BK \\ 0 & A - Lc \end{bmatrix} \right)^{-1} \begin{bmatrix} B \\ 0 \end{bmatrix} = [C \quad 0] \begin{bmatrix} sI_n - (A - Bk) & -BK \\ 0 & sI_n - (A - Lc) \end{bmatrix}^{-1} \begin{bmatrix} B \\ 0 \end{bmatrix}$$

$$= [C \quad 0] \begin{bmatrix} [sI_n - (A - BK)]^{-1} & * \\ 0 & [sI_n - (A - Lc)]^{-1} \end{bmatrix} \begin{bmatrix} B \\ 0 \end{bmatrix}$$

$$\frac{Y(s)}{R(s)} = C[sI_n - (A - BK)]^{-1}B$$

Output feedback control problem

- Note that the transfer function of the state feedback control system

$$\dot{x} = (A - BK)x + Br$$
$$y = Cx$$

is also

$$\frac{Y(s)}{R(s)} = C[sI_n - (A - BK)]^{-1}B$$

Thus, the ***output feedback transfer function*** will be ***unchanged*** if the state estimate $\hat{\boldsymbol{x}}$ is used for feedback instead of the true state $\boldsymbol{x}$.

- Compute the characteristic polynomial of the output feedback system

$$\begin{bmatrix}\dot{x}\\ \dot{\tilde{x}}\end{bmatrix}=\begin{bmatrix}A-BK & BK\\ 0 & A-LC\end{bmatrix}\begin{bmatrix}x\\ \tilde{x}\end{bmatrix}+\begin{bmatrix}B\\ 0\end{bmatrix}r\,,\qquad y=\begin{bmatrix}C & 0\end{bmatrix}\begin{bmatrix}x\\ \tilde{x}\end{bmatrix}$$

$$det\left(sI_{2n}-\begin{bmatrix}A-BK & BK\\ 0 & A-LC\end{bmatrix}\right)$$

$$=det\begin{bmatrix}sI_n-(A-BK) & -BK\\ 0 & sI_n-(A-LC)\end{bmatrix}$$

$$=det[sI_n-(A-BK)]\cdot det[sI_n-(A-LC)]$$

- The two terms above are called the ***separation property***, which allows us to ***calculate the feedback gain K and the observer matrix L independently*** as required if the system $(\boldsymbol{A},\boldsymbol{B},\boldsymbol{C})$is controllable and observable.

Closed-loop control design with prefilter

- State Space System

$$\dot{x} = Ax + B\,u$$
$$y = Cx$$

- Check observability and controllability

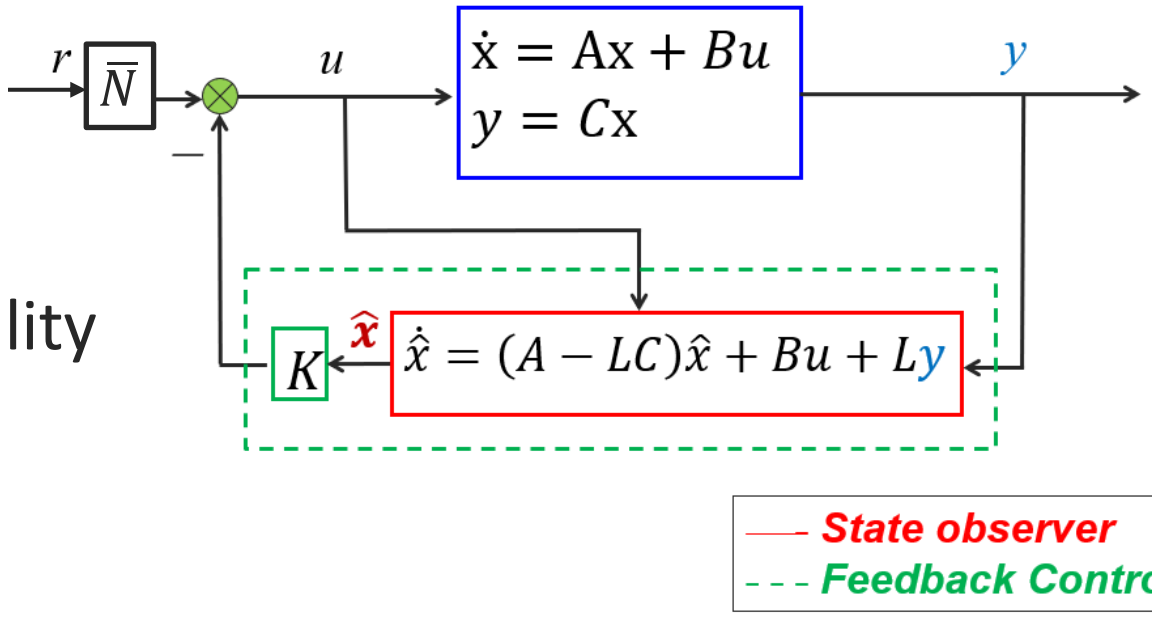
- Design a closed-loop observer

$$\dot{\hat{x}} = (A - L\,C)\hat{x} + Bu + L\,y$$

- Design a feedback control with prefilter

$$u = -K\hat{x} + \bar{N}r$$

where the feedback gain ***K*** *and the observer gain* ***L*** *are calculated independently.*



Example. Output feedback control problem

- **Example.** Consider a system described by the following transfer function

$$G(s) = \frac{1}{s(s+6)}$$

Find the feedback gain matrix $\boldsymbol{K}$, to place the closed-loop poles at $-4 \pm 6j$ and design a state observer of which the eigenvalues of $(\boldsymbol{A} - \boldsymbol{LC})$ are $-10, -10$.

- **Solution. Check:** The ***two-dimensional*** realization of the transfer function is ***controllable and observable***.

Example – Solution Find the feedback control gain K

- Write $G(s) = \dfrac{1}{s(s+6)}$ to state space.

Observable form will be used for the design.

$$\dot{x} = \begin{bmatrix} 0 & 0 \\ 1 & -6 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

Controllable form

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

- The desired closed-loop characteristic polynomial ($-4 \pm 6j$) is

$$(s+4+6j)(s+4-6j) = s^2 + 8s + 52$$

Example – Solution: find the control gain K

- Let $\boldsymbol{K} = [k_1, k_2]$, the closed-loop characteristic polynomial is

$$\det[s\mathrm{I} - (\mathrm{A} - BK)] = s^2 + (6 + k_1)s + 6k_1 + k_2$$

- Compute k_1, k_2 to match the desired polynomial $s^2 + 8s + 52$

$$(6 + k_1) = 8; \quad 6k_1 + k_2 = 52$$

$$k_1 = 2, k_2 = 40$$

- **Feedback control** $u = -K\hat{x} + \bar{N}\, r = -[2 \quad 40]\, \hat{x} + 52\, r$

- **Prefilter** $\bar{N}$, the closed-loop transfer function is

$$G_{cl} = \frac{1}{s^2+8s+52}, \bar{N} = \frac{1}{G_{cl}(0)} = 52$$

Example – Solution: find the observer gain L

- The required characteristic polynomial of observer $(-10, -10)$ is

$$(s+10)^2 = s^2 + 20s + 100$$

- Let $\boldsymbol{L} = [l_1, l_2]^T$, then the characteristic polynomial of $(\boldsymbol{A} - \boldsymbol{LC})$ is

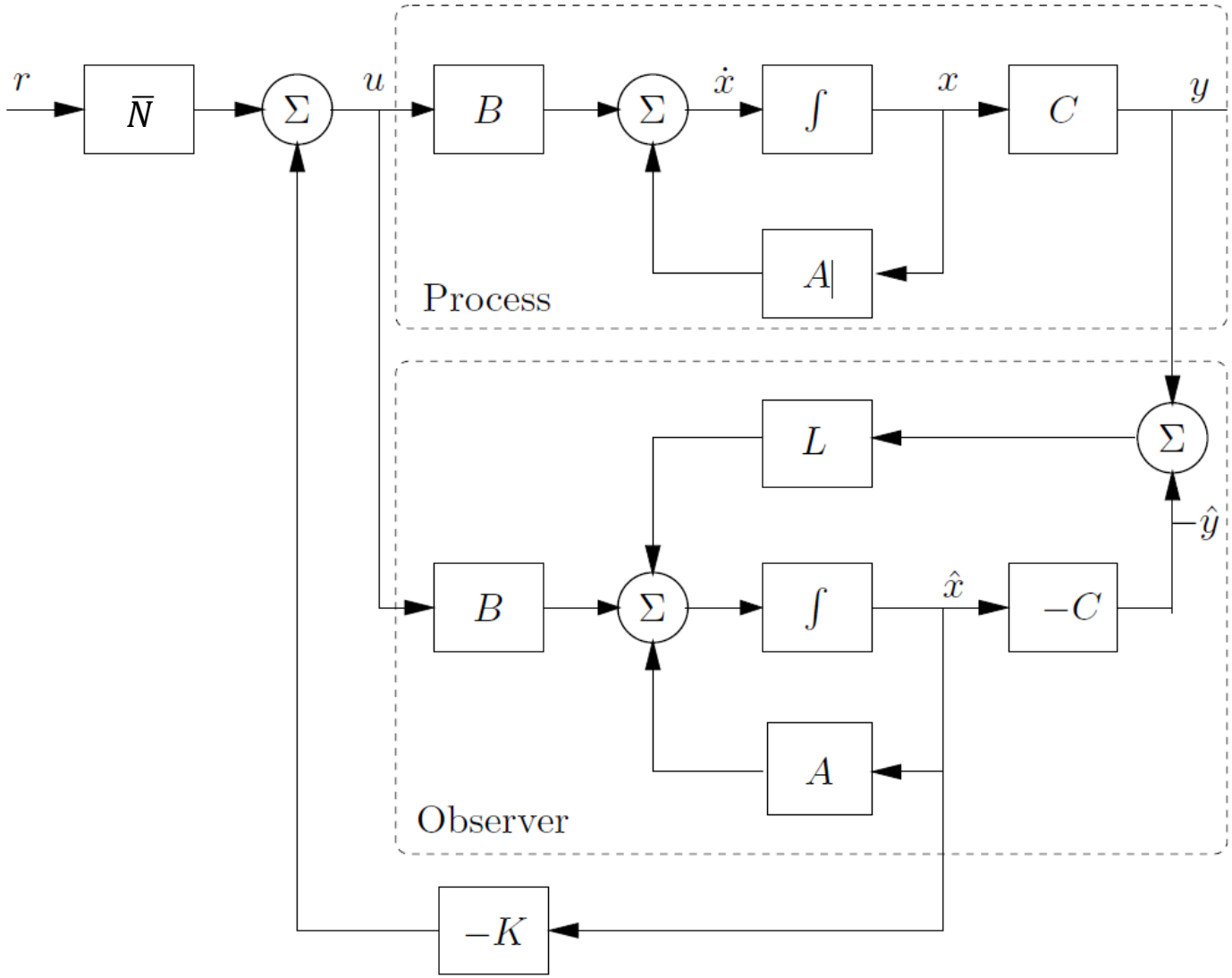
$$\det[s\mathrm{I} - (\mathrm{A} - LC)] = s^2 + (6 + l_2)s + l_1$$

$$l_1 = 100, l_2 = 14$$

- Thus, the state observer is designed as

$$\begin{aligned}\dot{\hat{x}} &= (A - LC)\hat{x} + Bu + Ly \\ &= \left(\begin{bmatrix}0 & 0\\ 1 & -6\end{bmatrix} - \begin{bmatrix}100\\ 14\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}\right)\hat{x} + \begin{bmatrix}1\\ 0\end{bmatrix}u + \begin{bmatrix}100\\ 14\end{bmatrix}y \\ &= \begin{bmatrix}0 & -100\\ 1 & -20\end{bmatrix}\hat{x} + \begin{bmatrix}1\\ 0\end{bmatrix}u + \begin{bmatrix}100\\ 14\end{bmatrix}y\end{aligned}$$

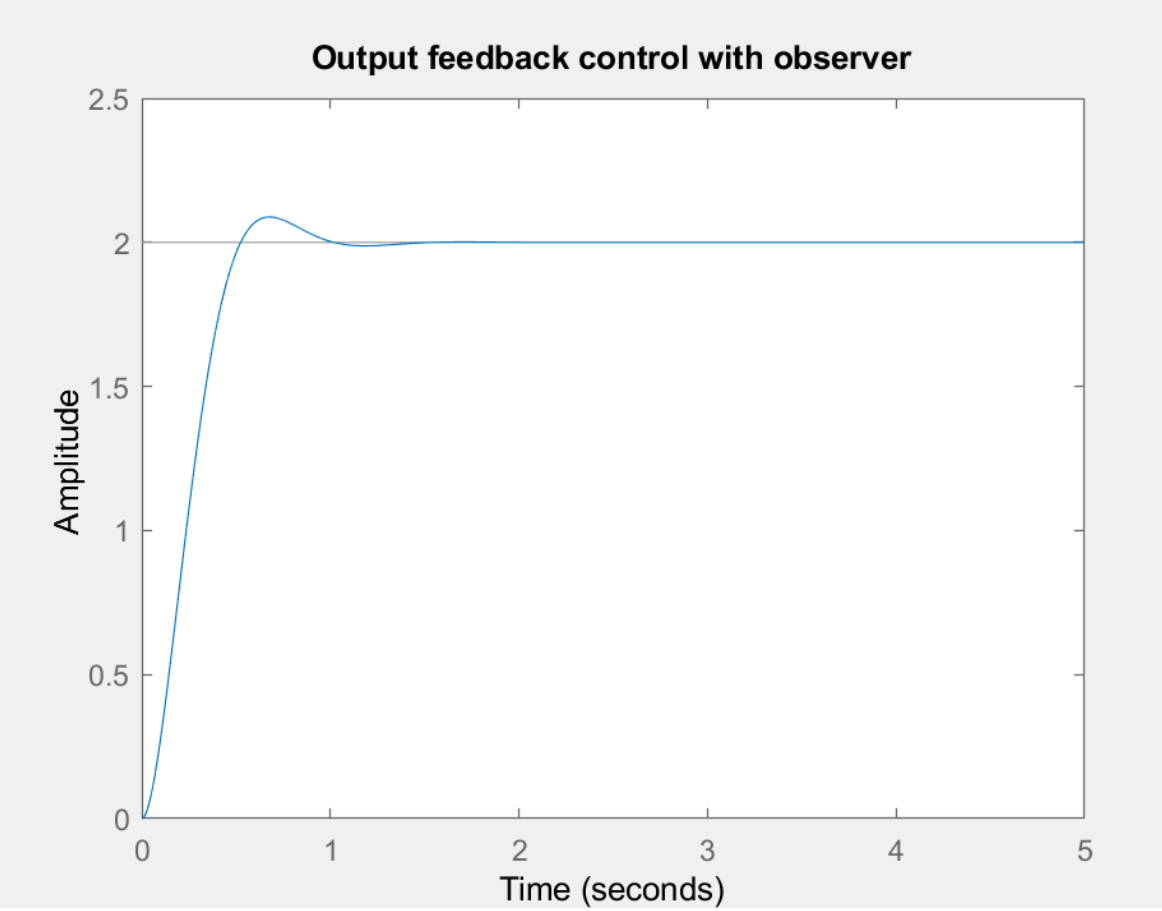
Output Feedback Tracking

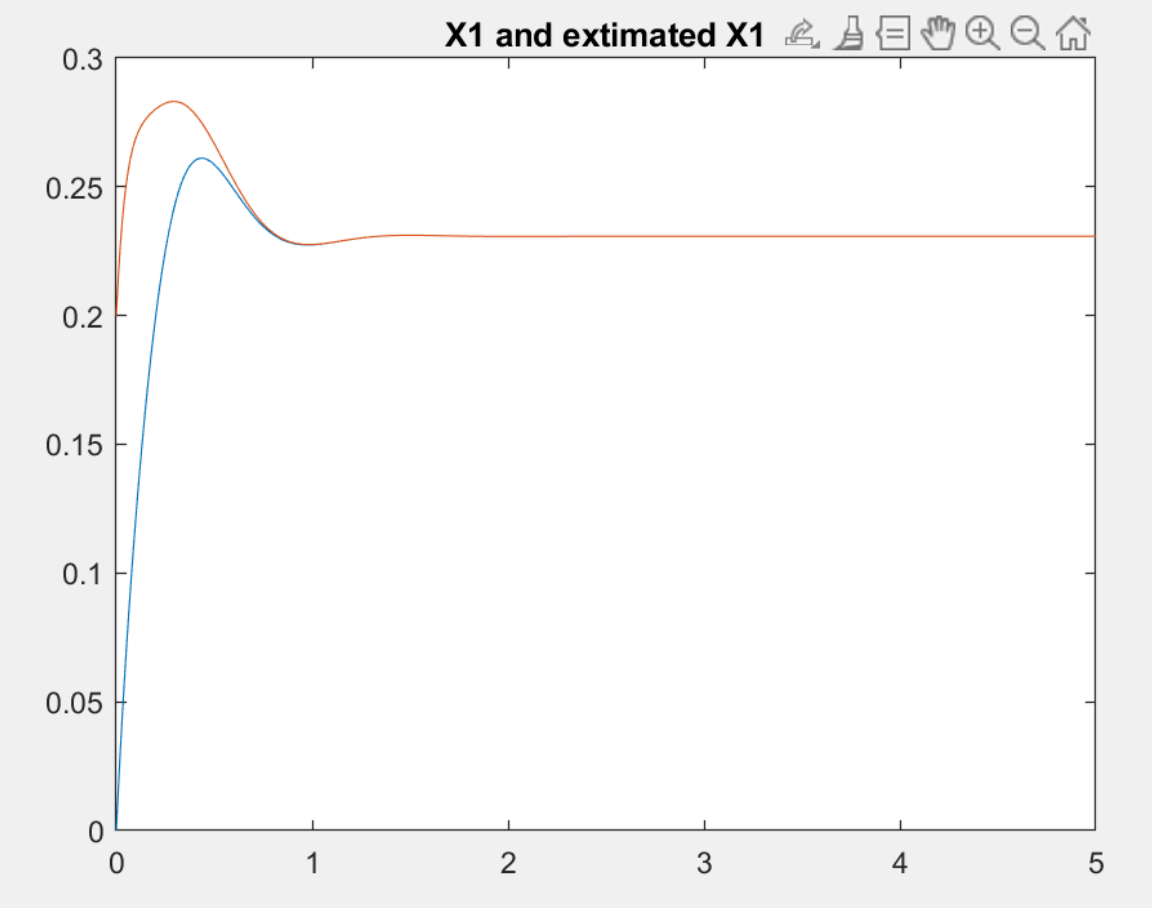


$$u = -K\hat{x} + \bar{N}\,r$$

$$\dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly$$

Example: Simulation with reference $r(t) = 2$





Matlab: L3f_ex_OutputFeedback.m

Takeaways

Benefits

- It allows us to calculate the feedback gain K and the observer matrix L independently as required if the system (A, B, C) is controllable and observable.

Drawbacks

- Need dynamic model of a system